Expert Systems With Applications 289 (2025) 128342

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Exploring biomedical relation extraction by combining natural language inference and dual dependency trees

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ARTICLE INFO

Keywords:
Biomedical relation extraction
Natural language inference
Dual dependency trees
Contrastive learning loss function
Voting decision mechanism

ABSTRACT

The task of relation extraction primarily focuses on predicting the relations between pairs of entities in text, which is crucial for applications such as drug discovery and disease diagnosis. However, this field faces two major challenges: First, traditional relation extraction models can utilize dependency trees to capture contextual information, but they still struggle to accurately identify the important semantic associations between entity words. Second, these models perform poorly when dealing with the complex structure of biomedical texts and domain-specific terms, particularly when applying cross-domain methods, which limits the models' ability to fully utilize textual information. We propose a novel approach that integrates Natural Language Inference by leveraging the textual entailment task to enhance the model's understanding and prediction of biomedical relations. This approach utilizes an improved contrastive learning loss function to better differentiate the entailment scores between relations. Additionally, a dual dependency tree is adopted to capture reliable dependency information and filter out noise by utilizing both inter-entity and intra-entity dependency connections and types. We further introduce an innovative voting decision mechanism to improve the model's ability to process and comprehend biomedical terms, thereby enhancing its performance on complex sentences. Experimental results show that our approach achieves state-of-the-art performance across three biomedical datasets, demonstrating its effectiveness and potential. Furthermore, ablation studies confirm the contribution of each technique, highlighting the importance of integrating NLI with dual dependency trees.

1. Introduction

Relation extraction (RE) aims to transform unstructured text into structured knowledge, enhancing the effectiveness of information extraction and supporting downstream tasks such as constructing knowledge graphs (Bosselut et al., 2019), natural language understanding (Wu et al., 2020), and recommendation systems (Zeng et al., 2020). In the biomedical field, relation extraction is primarily used to predict interactions between different molecules and biological molecules. For example, prediction of interactions between chemicals and proteins (Beltagy et al., 2019) can help identify new drug targets in the drug discovery and development process. In addition, this method is used to study drug-drug interactions (Segura-Bedmar et al., 2013), helping researchers better understand the complex dynamics of different drug combinations, thus improving medical safety and efficiency. Due to the complexity of biomedical knowledge and the high cost of manual curation, traditional biomedical relation extraction models have performed poorly in terms of efficiency and accuracy. As a result, researchers have gradually incorporated relevant domain technologies to improve the efficiency and accuracy of relation extraction, enabling models to automatically identify different biomedical entities and predict their interactions by learning features from large amounts of labeled data. The relation extraction task has garnered significant attention from researchers because of its ability to efficiently extract information from text. Typically, this task aims to predict the relation between every pair of entities in a given sentence. For example, in the sentence "$ER-\beta$ were down-regulated by either ethanol or melatonin". A relation extraction model needs to identify the two entities "ethanol" and "$ER-\beta$" and determine that the relation between these entities is "CPR:4". Therefore, accurate understanding of the context within the input content and determining the relations between entities in a sentence is crucial. However, there are currently three major challenges in the extraction of biomedical relations.

First, although biomedical literature websites have sufficient free biomedical literature and journal resources, these resources typically

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https://doi.org/10.1016/j.eswa.2025.128342
Received 29 September 2024; Received in revised form 8 May 2025; Accepted 24 May 2025
Available online 4 June 2025

do not include detailed expert annotations or in-depth analysis. Therefore, in these cases, one must rely on paid or high-level expert resources to gain a deeper understanding to organize them (Zeng et al., 2022) or rely on direct supervised learning (Qin et al., 2018). Consequently, Chen et al. (2022), Wei et al. (2020) proposed a method of using indirect supervision for relation extraction, which strengthens the target task with limited resources by using the supervisory signals of the source task with abundant resources. In this strategy, the training and inference processes of the target task are transformed to mimic the source task, thereby introducing additional supervisory signals that were originally inaccessible in the target task. In recent research, the combination of techniques from related fields to enhance the efficiency and accuracy of relation extraction has become a focus. Hu et al. (2022) have utilized Natural Language Inference (NLI) models for relation extraction. The NLI task aims to determine whether a hypothesis can be inferred from a given premise. NLI models learn generalized inductive biases of logical reasoning, which aligns closely with the goals of biomedical relation extraction. Therefore, we used NLI models to assist in the relation extraction model, utilizing the entailment task of NLI to handle relation extraction.

Secondly, while NLI models can accurately understand entities in biomedical domain sentences, the mentions of entities are often domain-specific terms that are rare in the pre-training corpus of language models. For example, in the ChemProt dataset (Beltagy et al., 2019), the relations are defined as interactions between gene complexes and other chemical substances; in the DDI dataset (Hu et al., 2022), the relations refer to the interactions between two drugs. Therefore, we found that in previous relation extraction models that used syntactic dependency trees, the relations provided long-distance context information, and syntactic information from dependency parsing automatically generated from the input sentences helped improve model performance. However, previous dependency relations generated from syntactic dependency trees focused primarily on the relations between entities, neglecting the dependency relations and types between entity words, which are also crucial for the relation extraction task. Based on this, we propose the construction of dual dependency relation trees that extract both the dependency relations between entities and those between entity words, providing additional contextual information for the NLI model.

Lastly, we noted that when dealing with unrelated entities, the predictive capability of NLI models is usually limited, a phenomenon particularly evident in the research on large language models (LLMs), such as in question-answering systems where LLMs tend to handle complex questions through multi-turn judgments (Sun et al., 2023). This approach inspired us to adopt a similar strategy to enhance the performance of NLI models. Thus, we propose to introduce a voting decision mechanism selector in our model. Specifically, when an NLI model faces a prediction task, instead of relying solely on the output of a single model, we use multiple models to make independent judgments on the same issue and decide the final output through a voting mechanism. This method effectively integrates the judgment power of multiple models, reducing the risk of bias or error from a single model. We have placed the code for the paper at: https://github.com/HUdoud/NLI_DDT_BIORE.
Our contributions to the relation extraction task are as follows:

- We have adopted cross-domain NLI technology and innovatively designed a new loss function with contrastive learning within the NLI model to expand the differentiation in entailment score rankings.
- We have utilized syntactic dependency tree technology to build a dual dependency tree, effectively compensating for the model's shortcomings in handling domain-specific nouns and contextual understanding.
- We have introduced a voting decision mechanism selector inspired by LLMs to determine the final predicted relation through voting.

2. Related works

2.1. Biomedical relation extraction

Relation extraction is a crucial subfield of Natural Language Processing (NLP) that aims to identify semantic relations between entities in text. In biomedical literature, relation extraction is particularly important as it can extract relations between entities such as drugs, diseases, and genes from a vast amount of scientific literature, thereby supporting clinical decision-making, disease research, and drug discovery. As the volume of medical literature continues to grow, biomedical relation extraction has become increasingly important. In sentence-level relation extraction research, the main challenge remains how to convert unstructured text data into a structured format that clearly expresses target entities and their relations. This is typically defined as a multi-class classification task, either by fine-tuning pre-trained language models or developing specialized pre-training objectives for relation extraction. For instance, researchers can boost the contextual representation methods of pre-trained language models by annotating subject and object entities. Li et al. (2019) further enhanced the performance of relation extraction by incorporating external knowledge from knowledge bases. Additionally, Tang et al. (2025) introduced a multi-scale attention mechanism, combining self-attention with depthwise separable convolution, to sharpen the focus on aspect and opinion terms.

In recent years, with the rapid development of LLMs, researchers have begun exploring their applications in relation to extraction tasks. For example, Kanjirangat et al. (2021) integrated a graph attention network with the T5 Transformer for relation extraction, aiming to identify drug-drug interactions and chemical-protein relations. Additionally, some researchers have reformulated relation extraction tasks using prompt learning (Chen et al., 2020; Ding et al., 2022). Additionally, prompt learning, as an emerging approach, has also been widely applied to the reformulation of relation extraction tasks. Ding et al. (2022) proposed a prompt-tuning method for relation extraction by constructing hierarchical prompts using logical rules. In the biomedical field, research on sentence-level relation extraction is relatively limited. El Khettari et al. (2024) proposed a two-stage information extraction system for document-level relation extraction, incorporating LLMs to address the challenge of ontological annotation in biomedical articles when entities are not explicitly mentioned. Despite the rapid advancement of LLMs, certain underlying challenges remain unresolved. For instance, LLMs may suffer from overthinking, which can impact result accuracy, and they may struggle with correctly interpreting entities that have multiple layers of meaning. In contrast, traditional BERT-based models demonstrate superior performance in relation extraction tasks, particularly in identifying implicit relationships. These studies not only drive advancements in relation to extraction technology but also provide strong support for knowledge discovery in the biomedical field. Despite this, traditional models have yet to fully address the challenges of sentence-level relation extraction, and several issues remain in the approach: (1) Although pre-trained language models demonstrate some capability in understanding context, they still fall short in deeply parsing specialized terminologies and complex structures. (2) The models continue to face significant challenges in judging and identifying implicit relations.

2.2. Natural language inference

Previous methods often lacked detailed descriptions of relationships, requiring explicit semantic modeling of relations during training. To address this issue, researchers have proposed a relation extraction method based on NLI. This approach provides the model with a premise and a hypothesis, predicting whether the facts in the premise necessarily entail those in the hypothesis, thereby enabling relation inference. Recently, Negru et al. (2025) introduced MorphNLI, a modular NLI approach based on text morphing, which significantly enhances the generalization ability of NLI tasks in cross-domain scenarios. Additionally, in the application of cross-domain models, researchers have explored various relation extraction methods based on natural language inference. researchers have proposed treating the relation extraction task as an autoregressive seq2seq task and using natural language inference models for extraction (Huguet et al., 2021). Other researchers have constructed detailed relation descriptions for each type of relation and used these descriptions to assess the accuracy of relation extraction through entailment relations (Ma et al., 2021; Sainz et al., 2021). Hu et al. (2022) further proposed a natural language inference-based relation extraction method, which models relation extraction as a natural language inference task, making it more akin to a multiple-choice question rather than a traditional classification task. By explicitly providing relation descriptions and incorporating prior knowledge, this method enhances the model's understanding of relation semantics, thereby improving the accuracy of relation extraction. Although natural language inference-based methods have made significant progress in relation extraction, they still have certain limitations in evaluating the accuracy of inferred relations. For example, these models struggle with complex semantic relationships and long-distance dependencies, and their adaptability to domain-specific knowledge remains limited.

2.3. Dual dependency trees

To enhance contextual understanding in relation extraction tasks, previous research has extensively explored the application of neural networks, including the use of CNNs (Xu et al., 2016; Zeng et al., 2014), RNNs (Socher et al., 2012; Zhou et al., 2016), BERT encoders (Baldini Soares et al., 2019; Wang et al., 2019; Wu et al., 2018) and various BERT variants (Markus et al., 2020; Wu et al., 2019). Furthermore, to improve model performance, research has shown that incorporating additional knowledge into models can significantly enhance results. This knowledge typically includes lexical, syntactic, and semantic knowledge, with syntactic dependencies being particularly important in this task (Xu et al., 2015). Moreover, Fan et al. (2025) designed a Forest-Induced Tree Enhanced Graph Attention Network based on the integration of syntactic structures and Graph Neural Networks. This model utilizes a graph attention network to assign weights to dependency edges under syntactic constraints, thereby preventing the propagation of irrelevant word information. Some researchers later attempted to integrate automatically generated dependency syntax analysis knowledge of sentences into models (Guo et al., 2019a; Mandya et al., 2020; Sun et al., 2020). Zhang et al. (2024a) adopted a soft pruning method to compute key information and executed it in parallel with contextual encoding, thereby improving model performance. Park et al. (2024) in studying traditional dependency tree methods, explored the trade-off between information loss and syntactic weakening and proposed a relation extraction model based on an Entity-Centered Dependency Tree. Although Huang et al. (2025) proposed a multi-view attention syntax-enhanced graph convolutional network that leverages an attention mechanism to weight the information from different syntactic views, existing syntactic knowledge modeling methods remain limited to a single level, making it difficult to fully capture deep relations and semantic information between entity words. Furthermore, these methods may suffer from insufficient information propagation when dealing with complex sentence structures or long-distance dependencies, thereby affecting the model's ability to understand and extract implicit relationships. Therefore, an important challenge in current research is how to more effectively model multi-level syntactic knowledge and enhance the model's ability to capture long-distance dependencies.

3. Method

In this section, we will describe the structural framework of the proposed model in detail, as illustrated in Fig. 1.

3.1. Task restatement

The models for the relation extraction task are primarily trained on input sentence $\mathcal{S}$ that contain entities E_n, where n is the number of entities in the sentence, and $n \in \{1, 2\}$. Within these entities, each word W_{n_i} forms the basic components of both the entity and the sentence. Here, n_i is the index of the word W_{n_i} within $\mathcal{S}$, with $i \in \{1, 2, \cdots\}$. The core objective of the training is to accurately predict the relations χ between entities E_n in the dataset $\mathcal{D}$, based on given relation labels $\mathcal{C}$, where $\chi \in \mathcal{C}$.

3.2. Relation extraction combined with natural language inference

Building on the research approach of Xu et al. (2023), we combine NLI tasks with relation extraction tasks to leverage the advantages of NLI tasks to enhance the accuracy of relation extraction. The model diagram is shown in Fig. 1. (a). By utilizing models trained in NLI tasks

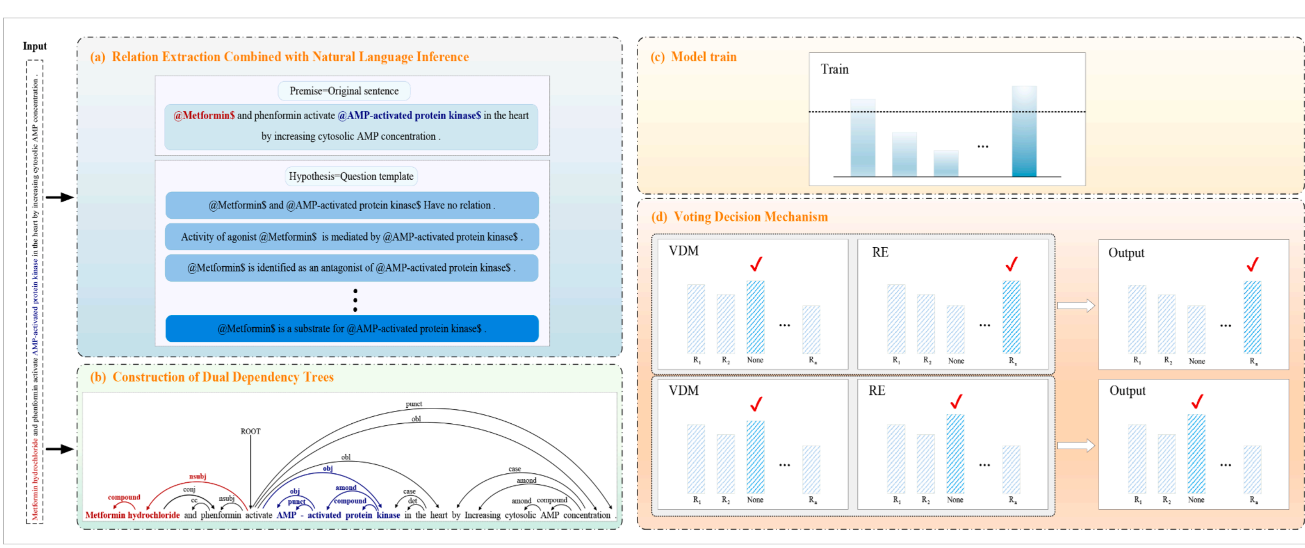

Fig. 1. The model diagram in this paper illustrates the following: (a) is the relation extraction module combined with natural language inference, and (b) is the dual dependency trees construction module. The outputs of (a) and (b) are combined and fed into (c) for training. Finally, the voting decision mechanism selector in (d) is used to vote on the final output. If both the VDM and the relation extraction model predict the instance as "no relation", the final result is labeled "no relation". If either model predicts a relation, the result will return the prediction from the relation extraction model.

to support relation extraction tasks, NLI models primarily determine the semantic relation between two sentences. These relations are typically divided into three types: Entailment, Contradiction, and Neutral. Within the NLI model framework, we specifically employ its textual entailment task techniques. The textual entailment task is based on a given premise text, inferring the relation of a related hypothesis text with the premise. Therefore, we decompose the relation extraction task into multiple NLI query tasks, where the NLI model receives a premise and a hypothesis and outputs the corresponding relation type (including no relation, denoted by a label of 0 or None).

We decompose it into C hypothesis sets based on the number of relation labels C in the dataset, each hypothesis query corresponding to a relation χ. We use the input sentence $\mathcal{S}$ of the relation extraction model as the premise for the NLI model and a template sentence describing the relation χ between the two mentioned entities as the hypothesis.

Firstly, regarding the setup of hypothesis templates, we found that in the previous research method by Xu et al. (2023), the entities considered are mostly domain-specific terms, which rarely appear in the pre-trained corpora of language models, leading to insufficient training on rare biomedical terms. To address this issue, we propose using dual dependency trees to overcome this difficulty (which will be detailed in Section 3.3). We maintain the original structure of entities without masking, integrating them with a dual dependency tree for relation extraction through natural language inference. Secondly, although complex templates that contain excessive details enrich the description, they can also introduce unnecessary or irrelevant information, potentially confusing the model. Furthermore, using highly detailed and specific templates may cause the model to overfit the training data. Therefore, for each relation χ, we set it as the most straightforward descriptive natural language template: Combining contextual information, the relation between two entities is articulated using the phrase "is", where the natural language hypothesis template is defined as $P(\chi)$.

Secondly, in inferring the relations between entities using natural language inference, we calculate the confidence scores for the corresponding relations and determine the logical values for all corresponding hypothesis possibilities. The calculations involve combining the sentence, entities, and hypotheses. For the entities E_n contained in the given input sentence $\mathcal{S}$ and each relation label $\chi \in C$, the confidence scores are calculated using the following Eq. (1).

$$g(\chi) = f_{NLI}\left(\mathcal{S}[\text{SEP}]E_n[\text{SEP}]P(\chi)\right) \tag{1}$$

The marker $[SEP]$ is a delimiter used to separate the sentence, entities, and hypotheses.

Finally, during the training phase, we use the InfoNCE loss function from contrastive learning to increase the separation between positive and negative examples. In each step, m negative sample relations $\{\chi_1, \chi_2, \cdots, \chi_m\} \in C$ are extracted and $g_{\chi_1}, g_{\chi_2}, \cdots g_{\chi_m}$ are calculated. Combined with focal loss, the weight of the loss for hard-to-classify samples is increased, improving the ranking of the entailment scores for the correct relation labels. Specifically, the InfoNCE loss we optimize is shown in the following Eqs. (2) and (3).

$$H_t = \frac{\exp(g(\chi)/\tau)}{\exp(g(\chi)/\tau) + \sum_{i=1}^{m} \exp(g(\chi_i)/\tau)} \tag{2}$$

$$L_{FLNCE} = -\left(1 - H_t\right)^{\gamma} \Sigma_{(S,\chi)} \log H_t \tag{3}$$

We rank the entailment scores for each hypothesis and make the final prediction based on the relation corresponding to the highest-scoring hypothesis.

3.3. Construction of dual dependency trees

In response to the issue raised in Section 3.2, as shown in Fig. 1.(b), we designed dual dependency trees. We observe that entities in English sentences are often compound entities. For example, "metformin hydrochloride" is composed of two component words: "metformin" and "hydrochloride". When applying dependency parsing tools to analyze such sentences, there are typically clear dependency relations among the words that make up these entities. Based on this observation, we propose a dual dependency tree structure. This structure not only captures the dependency relations among the component words within a single entity but also models the interactions between each component word of one entity and all the component words of another entity, thereby providing the model with richer semantic information. As illustrated in Fig. 2, this approach enables more effective capture of semantic relations between cross-lingual entities and enhances the model's understanding of sentence structure. For instance, the compound entity "metformin hydrochloride" is decomposed into its components "metformin" and "hydrochloride", and the dependencies between them form the intra-entity dependency. At the same time, in the inter-entity dependency, the model is able to accurately identify the dependency paths between "metformin" and each component of another entity, "AMP-activated protein kinase", thereby revealing the potential semantic connections across different entities.

To construct dual dependencies, we used an existing dependency generator (Stanford CoreNLP) to generate the dependency analysis results for the input sentence $\mathcal{S}$.In the generated dependency tree, each word in $\mathcal{S}$ is connected to its governor and dependent through their respective dependency relations, with a corresponding path existing between any two words in $\mathcal{S}$. In the dependencies, we considered two types of dependency information and constructed corresponding dependency path sentences and a dependency type matrix to provide supplementary information for relation extraction. The first type of dependency is the intra-entity dependency, also referred to as "intra-entity" dependency, which is formed by the dependency paths between each word W_{n_i} within the entity E_n. The second type is the extra-entity dependency, also known as "cross-entity" dependency, constructed from the vocabulary of the entity and dependency paths, following the dependency paths between the word W_{n_i} in $\mathcal{S}$ and words of other entities.

Fig. 3 illustrates the process of constructing dual dependencies from the dependency tree of a sentence. Next, we will explain in detail how to construct dual dependencies.

"Intra-entity" Dependence

"Intra-entity" dependency refers to the dependency paths among the words that constitute an entity, providing more comprehensive information about the relations between the words within the entity. To construct "intra-entity" dependencies, we first identify each word Wni that constitutes the entity from the dependency tree generated by the dependency parser, noting the governor and dependent in $\mathcal{S}$ or each word W_{n_i}. Next, we use the governor and dependent of the word W_{n_i} in $\mathcal{S}$ as the keys for the "intra-entity" dependency, and their respective relations with W_{n_i} as the values for these keys, thus obtaining a corresponding key-value pair. If there are multiple dependency relations, this forms a group of key-value pairs. The key-value pair is represented as $X_{n_i}^{\text{inside}} = \left(K_{n_i}^{\text{inside}}, V_{n_i}^{\text{inside}}\right)$, where $K_{n_i}^{\text{inside}}$ is the word connected to the entity word W_{n_i} through a dependency relation, and $V_{n_i}^{\text{inside}}$ is the type of dependency relation between the word connected to the entity word W_{n_i}. For example, in the first entity phrase "metformin hydrochloride" shown in Fig. 3, we select the word "metformin" (highlighted with a yellow background in the figure) and construct its intra-entity dependency path. According to the dependency tree, there is a dependency relation between "metformin" and "hydrochloride", with the dependency type labeled as compound. Therefore, the intra-entity dependency path is: " metformin → hydrochloride " , with the dependency type compound. Thus, obtaining the corresponding key-value pair as shown in Eq. (4).

$$X_{1_1}^{\text{inside}} = \left[\left(\text{hydrochloride}, \text{compound}\right)\right] \tag{4}$$

Next, the word "hydrochloride" in the first entity word "metformin hydrochloride" is associated with the words "metformin", "phenformin", and "activate". Therefore, based on their respective dependency rela-

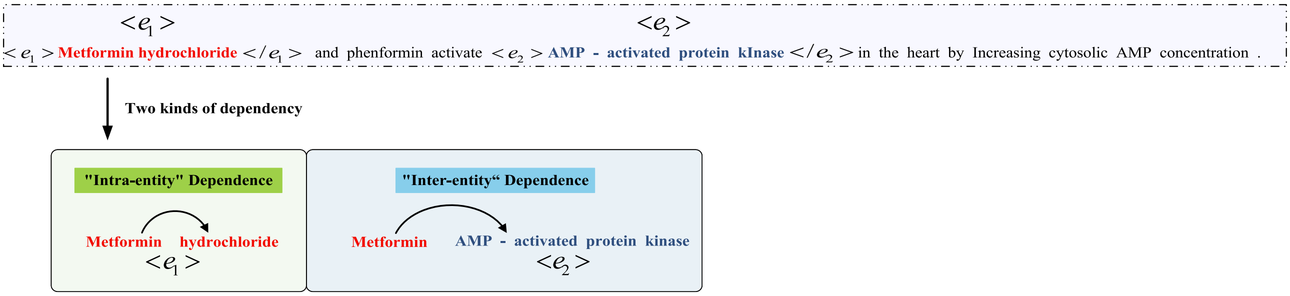

Fig. 2. Example of dual dependency relations.

tion types, we can derive the corresponding group of key-value pairs as shown in Eq. (5).

$$X_{1_2}^{\text{inside}} = \big[(\text{metformin}, \text{compound}), (\text{phenformin}, \text{conj}), (\text{activate}, \text{nsubj})\big] \tag{5}$$

Based on the critical pairs of "intra-entity" dependence, we can obtain the corresponding dependency paths, dependency path matrix, and dependency type matrix. The dependency path refers to the path from a word W_{n_i} within the entity E_n to the word where it ultimately ends. The dependency path matrix is generated based on these dependency paths, and the dependency type matrix is generated according to the corresponding types of dependencies in the dependency tree within the paths. Using the above method, we obtained the dependency relations, dependency type sentences, and dependency path matrix for the "intra-entity" dependencies.

"Inter-entity" Dependence

"Inter-entity" dependence refers to the dependency paths between two entities, providing more comprehensive contextual information based on their dependency paths. Before constructing "inter-entity" dependence, we considered that in the structure of noun phrases in English, the head noun (i.e., the main component of the noun phrase) typically appears at the end of the phrase. Therefore, when dealing with dependencies between entities, the reference point used is the last word of the entity. Based on the previously generated dependency tree, we locate the first entity word E_n and the second entity word $E_{n'}$. First, we identify the dependency path from the last word of entity E_n to the last word of entity $E_{n'}$. Then, similar to the process of constructing "intra-entity" dependence described above, we extract all the words and their dependency relation types from the dependency parse tree along the path connecting the two entities. We take the extracted words as keys and the dependency relation types between them as values, obtaining the key-value pairs for "inter-entity" dependence. The representation of these key-value pairs is shown in Eq. (6).

$$X_{n_i}^{\text{outside}} = \left(K_{n_i}^{\text{outside}}, V_{n_i}^{\text{outside}}\right) \tag{6}$$

where $K_{n_i}^{\text{outside}}$ refers to the word connected to the entity word E_n through a dependency relation, and $V_{n_i}^{\text{outside}}$ refers to the dependency relations between them. For example, in Fig. 3, the dependency path between the word "metformin" in the first entity "metformin hydrochloride" and the last word "kinase" in the second entity "AMP-activated protein kinase" can be found as follows: metformin → hydrochloride → activate → AMP → – → activated → protein → kinase, And their corresponding dependency relation types are: compound, nsubj, obj, amod, punct. Therefore, we can derive the key-value pair set for the "inter-entity" dependence, as shown in Eq. (7).

$$X_{1_1}^{\text{outside}} = \big[(\text{hydrochloride}, \text{compound}), (\text{activate}, \text{nsubj}), (\text{kinase}, \text{obj}), (\text{activated}, \text{amod}), (-, \text{punct}), (\text{protein}, \text{compound}), (\text{AMP}, \text{obj})\big] \tag{7}$$

The method for obtaining the dependency type matrix and dependency path matrix for "inter-entity" dependence is the same as that for obtaining the dependency type matrix and dependency path matrix for "intra-entity" dependence. Therefore, following the steps above, we can obtain the corresponding dependency relations, dependency type matrix, and dependency path matrix for "inter-entity" dependence. Through these two methods, we complete the construction of the dual dependency trees.

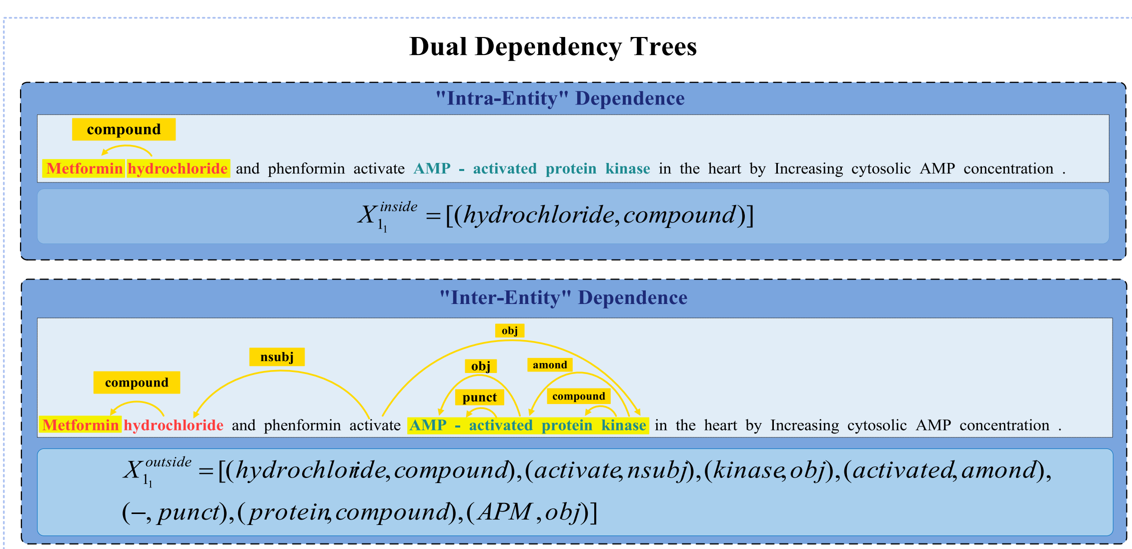

Fig. 3. Construction of the dual dependency trees.

Synergistic mechanism between NLI and dual dependency trees

To enhance semantic understanding in relation extraction, we integrate dual dependency trees with a NLI model through a synergistic mechanism. The dual dependency trees are designed to capture syntactic dependencies at both the intra-entity and inter-entity levels. These dependencies are encoded into two structural representations: a dependency path matrix and a dependency type matrix.

These syntactic features are then concatenated with the semantic embeddings generated by the NLI model. In particular, the NLI model computes entailment scores by analyzing premise-hypothesis pairs, providing a measure of logical consistency between textual elements. Meanwhile, the syntactic matrices serve as structural constraints that guide and refine these entailment scores. For example, in the hypothesis "Metformin activates AMP-activated protein kinase", the inter-entity dependency path, such as metformin → activate → kinase-extracted from the dependency tree provides explicit syntactic evidence of the action-entity relationship. This syntactic signal, encoded in the dependency matrices, reinforces the NLI model's ability to align semantic roles and improve inference quality.

3.4. Voting decision mechanism

In recent research on relation extraction models, we explored the issue of accuracy in relation extraction models when dealing with unlabeled data and found that the models have lower accuracy in determining unlabeled instances. Inspired by the negotiation strategy proposed in LLMs research (Sun et al., 2023), we attempted to simplify the relation extraction task into a binary classification problem, with the labels categorized as "labeled" and "unlabeled (None)". We trained the model on the Voting Decision Mechanism (VDM) and then designed a voting-based decision mechanism, as shown in Fig. 1. (d). For each test instance, we first calculate the scores for the relations, then rank them according to the scores, and determine the final prediction result through a voting mechanism.

To more accurately determine the final prediction results, we have introduced a weighted fusion scoring mechanism before the VDM, which allows for dynamic adjustments based on the distribution of weights. By adjusting the model scores, a process for dynamically adjusting voting weights has been incorporated. The implementation of dynamic adjustment of voting weights is shown in Eq. (8).

$$w_r = \frac{\exp(\eta \cdot \text{Accuracy}_r)}{\sum_{j=1}^{R} \exp(\eta \cdot \text{Accuracy}_j)} \tag{8}$$

Among them, $Accuracy_r$ is the accuracy of the r-th round on the validation set, and η is the smoothing coefficient. For the smoothing coefficient, we adopted methods proposed by Hinton et al. (2015), Dietterich (2000), and Yang et al. (2023b). When the value of η is large, the weight becomes more sensitive to the accuracy differences across rounds, allowing voting rounds with better validation performance to receive higher weights. In contrast, a smaller η leads to a flatter weight distribution, resulting in more uniform weighting across rounds. By adjusting η, the model can dynamically emphasize more reliable rounds while maintaining a certain degree of flexibility. In addition, we have adjusted the scoring calculation $\mathrm{g}(\chi)$ for the model. Assuming there are r rounds of voting, the score for the corresponding relation χ is $\mathrm{g}_r(\chi)$, and the final weighted fused score is given by Eq. (9).

$$g(\chi) = \sum_{r=1}^{R} w_r \cdot g_r(\chi) \tag{9}$$

where w_r is the voting weight of the r-th round. At the beginning, R is small, and the weight differences across rounds are small, preserving diversity to avoid overfitting. As training progresses, R increases, and the weight of the accuracy rounds is significantly enhanced by the effect of η, forming an adaptive focusing mechanism. This process does not require adjusting R, but rather achieves dynamic adjustment of the weight distribution naturally through η.

The VDM model is essentially the same as the model we trained earlier, but the way we handle prediction classification has been adjusted. We introduced a class score regularization mechanism that calculates the scores for only two categories (i.e., "relation present" and "no relation") to improve the model's accuracy in predicting these two categories. This ensures that instances with a clear relation are not mistakenly classified as "no relation" and also prevents indeed "no relation" instances from being incorrectly classified as "relation present". When calculating the class score regularization, we added a margin algorithm to ensure that the model ranks correctly and makes the score difference between correct and incorrect labels sufficiently large. The specific formulation is shown in Eq. (10).

$$\ell_{\text{ord}}(g(\chi), g(\chi'), y) = \max\left(0, y - \left(\sum_{r=1}^{R} w_r \cdot g_r(\chi) - \sum_{r=1}^{R} w_r \cdot g_r(\chi')\right)\right) \tag{10}$$

Here $\mathrm{g}(\chi)$ and $\mathrm{g}(\chi_i)$ are the scores calculated by the model for two relations. γ is a positive number representing the margin size. The function max ensures that the loss value is non-negative. When the score of $\mathrm{g}(\chi)$ is higher than that of $\mathrm{g}(\chi_i)$ and the difference is greater than or equal to γ, the loss is zero. This means that the model has successfully ranked $\mathrm{g}(\chi)$ ahead of $\mathrm{g}(\chi_i)$ and met the margin requirement. When the difference between the scores of $\mathrm{g}(\chi)$ and $\mathrm{g}(\chi_i)$ is less than γ, the loss is positive. This indicates that the model has not successfully ranked $\mathrm{g}(\chi)$ ahead of $\mathrm{g}(\chi_i)$ or that the difference has not met the margin requirement. By introducing this ranking loss, the model is trained to pay more attention to distinguishing the scores of different relations or instances, ensuring that the correct relation score is significantly higher than the incorrect one. On this basis, the class score regularization mechanism strategy is formulated as shown in Eqs. (11) and (12), where $\mathcal{W}$ is the voting weight, and $\mathcal{W} = (w_1, w_2, \ldots, w_r)$.

$$\mathcal{L}_{\text{VDM}} = \sum_{(\text{sencent},\chi)\in D} \ell_{\text{VDM}}(\mathrm{S}, \chi; \mathcal{W}) \tag{11}$$

$$\ell_{VDM} \triangleq \begin{cases} \sum_{i=1}^{m} \ell_{\text{ord}}\left(\sum_{r=1}^{R} w_r \cdot g_r(\chi), \sum_{r=1}^{R} w_r \cdot g_r(\chi_i), \gamma\right), & \chi = \text{None} \\ \ell_{\text{ord}}\left(\sum_{r=1}^{R} w_r \cdot g_r(\chi), \sum_{r=1}^{R} w_r \cdot g_r(\chi_i), \gamma\right), & \chi = \text{label} \end{cases} \tag{12}$$

Through the above methods, after training, the VDM is used as a discriminative tool to identify relation types, including "no relation" and specific relation categories. We compared and combined the VDM model with the previously trained relation extraction model, forming a two-stage prediction process as follows:

- **First round prediction:** The VDM model is used to make an initial prediction for each instance to determine whether the instance belongs to the "no relation" category or a relation category.
- **Decision logic:** If the VDM predicts "no relation", this prediction is directly accepted, and no further processing is carried out. If the VDM predicts a relation, the instance is passed to the relation extraction model, which makes a more detailed prediction of the relation category.
- **Voting strategy:** In the final prediction decision, the instance is only marked as "no relation" if both the VDM and the relation extraction model predict "no relation". If either of the two models predicts a relation, we return the prediction result from the relation extraction model.

Table 1
Statistics of datasets.

Dataset	Train	Test	Dev	Relation
DDI	25,296	5716	2496	5
ChemProt	18,035	15,745	11,268	6
GAD	4261	534	535	2

4. Experiment

4.1. Data

This paper primarily evaluates and analyzes the model on three biomedical datasets, as shown in Table 1: Drug-Drug Interaction, ChemProt, and GAD. Table 1 shows the statistical information for each dataset.

DDI: The DDI dataset studies drug-drug interactions, focusing on pharmacovigilance constructed from PubMed abstracts. It defines five types of drug entity interaction relations: advice, effect, interaction, mechanism, and negative.

ChemProt: The ChemProt dataset consists of a PubMed abstract corpus annotated with five high-level chemical-protein interaction types, including six relation types ("CPR:3", "CPR:4", "CPR:5", "CPR:6", "CPR:9", and "None").

GAD: The GAD dataset is a semi-annotated dataset created using the Genetic Association Database. It contains gene-disease associations. It has only two relation types ("1" indicating related and "0" indicating unrelated).

We use the micro-average F1 score to calculate the performance of all instances as our evaluation metric.

4.2. Baseline

We compared our proposed model with various baseline methods and categorized these baselines into the following classes based on their model frameworks.

BERT-based biomedical models: BioBERT (Lee et al., 2020) is based on the original BERT model developed by Google, pre-trained on a large corpus of text to achieve deep bidirectional representation; PubMed-BERT (Gu et al., 2021) is a fine-tuned model released by Tinn et al. (2023), pre-trained on the aforementioned corpus; BLUE-BERT (Peng et al., 2019) is a BERT-based model designed explicitly for biomedical and clinical text processing; KeBioLM (Yuan et al., 2021) is an improved version of PubMedBERT and uses the BERT architecture as its core structure; BioLink-BERT (Yasunaga et al., 2022) introduces a pre-training task for link prediction, enabling the model to learn multi-hop knowledge.

Architecture-specific biomedical models: Sci-BERT (Beltagy et al., 2019) is a BERT model pre-trained on a scientific corpus, which includes 1,140,000 full-text papers from Semantic Scholar; BioRE-Prompt (Yeh et al., 2022) is initialized from RoBERTa trained on Semantic Scholar, utilizing a tri-mark prompt learned for each relation. Inference is conducted by identifying the best matching prompt; BioMegatron (Shin et al., 2020) is a Transformer model optimized for large-scale training tasks, pre-trained on a commercial subset of PMC; FPE (Zhu et al., 2024) is a novel joint extraction method called Feature Partition Encoding proposed for the biomedical entity and relation joint extraction task, which addresses the issues of task interaction imbalance and feature sharing; HKG (Asada et al., 2023) is a joint model that integrates textual information with heterogeneous pharmaceutical knowledge graphs; BioE-GRE (Zheng et al., 2023) syntactic topology through graph structures, combining pre-trained language models and graph neural networks to capture complex syntactic relationships in biomedical texts explicitly.

Large language model-based biomedical models: Significant advancements have been made in the study of LLMs in the field of biomedical natural language processing. BioGPT (Luo et al., 2022) is based on the GPT-2 architecture, pre-trained on 15 million PubMed abstracts, aimed at addressing the limitations of existing BERT-based models (such as BioBERT and PubMedBERT) in biomedical text generation; SciFive (Phan et al., 2021) is a text generation model based on the T5 architecture, specifically designed for natural language processing tasks in the biomedical domain; Chen et al. (2025) conducted a systematic examination of the performance and application potential of GPT-3.5 and GPT-4 in biomedical relation extraction tasks, providing important references for the application of LLMs in this field. Building on this, Zhang et al. (2024b) further expanded the scope of research, providing a comprehensive summary and in-depth analysis of the application of various large language models, including GPT-3.5-turbo, in the field of relation extraction.

Task-specific optimized models: BioM-ALBERT (Alrowili et al., 2021) is further pre-trained based on ALBERT using a large volume of biomedical literature; TaMM-BERT (Chen et al., 2021) is a word dependency relation extraction model; NBR (Xu et al., 2023) is a relation extraction model in the NLI domain; BioM-ELECTRA (Alrowili et al., 2021) is a pre-trained language model optimized for biomedical text processing tasks based on Google's ELECTRA model framework; SPBERE (Yang et al., 2023a) explores a fundamental task in biomedical text mining: triplet extraction. It introduces a new pipeline approach to address the challenges of entity and relation extraction in the biomedical field, particularly the detection of overlapping entities and relations; HEFed (Wang et al., 2025) reduces communication overhead through knowledge distillation and enhances model convergence by maximizing mutual information, thus achieving efficient privacy preservation in the task of medical relation extraction; WGCN (Wang et al., 2024) analyzes the syntactic dependencies at the sentence level to construct adjacency, dependency, and weight matrices, thus distinguishing the importance of different dependencies for relation extraction, while employing a pruning strategy to eliminate low-weight dependencies and suppress noise.

4.3. Experimental results

The experimental results of our model are shown in Table 2. We compared the results of existing model methods with those of our model, and we found that our results surpass the best results of the cross-domain model NBR (Xu et al., 2023) by the following margins: 1.92 % on the DDI dataset, 0.74 % on the ChemProt dataset, and 1.01 % on the GAD dataset. Moreover, our proposed cross-domain relation extraction model not only surpasses traditional dependency tree models such as TAMM-BERT $_{\text{large}}$, but also outperforms LLMs like GPT-3 and GPT-4 in terms of performance.

Firstly, it is evident that in previous models that did not utilize cross-domain models, our model combined with the NLI model, which focuses on understanding the logical relations between sentences, helps the model better understand and distinguish the complex semantic relations between entities, thereby improving the accuracy of relation classification. Even when faced with complex or ambiguous texts, the NLI model we constructed with the contrastive learning loss function can infer the most likely relation type through its reasoning capabilities, even in the absence of explicit expressions.

Secondly, integrating syntactic dependency relations with the NLI model for relation extraction aids the model in understanding the structural relations between words in a sentence, accurately locating the connections between keywords and entities, and capturing the deep semantic information of the sentence. This integration enhances both the model's performance and accuracy.

Finally, we have noticed that our model performs better compared to LLMs. This may be because LLMs tend to overanalyze when dealing with rare entities in biomedical texts, and the lack of sufficient contextual information in the sentences leads to less precise judgments. Additionally, our model demonstrates excellent generalization capabilities, making multiple assessments on categories without relationships to ensure

Table 2
Experimental results.

Method	Model	DDI	ChemProt	GAD
BERT-based biomedical models	BLUE-BERT$_{large}$ (Peng et al., 2019)	79.90	74.40	-
	BioBERT$_{large}$ (Lee et al., 2020)	80.33	76.46	79.83
	BioBERT$_{base}$ (Lee et al., 2020)	81.32	74.93	-
	KeBioLM (Yuan et al., 2021)	82.04	79.30	-
	PubMed-BERT$_{base}$ (Gu et al., 2021)	82.36	77.24	82.34
	BioLink-BERT$_{base}$ (Yasunaga et al., 2022)	82.72	77.57	84.39
	BioLink-BERT$_{large}$ (Yasunaga et al., 2022)	83.35	79.98	84.90
Architecture-specific biomedical models	BioRE-Prompt (Yeh et al., 2022)	-	67.46	-
	BioMegatron (Shin et al., 2020)	-	70.00	-
	BioEGRE (Zheng et al., 2023)	-	79.97	83.31
	FPE (Zhu et al., 2024)	81.70	-	-
	Sci-BERT$_{base}$ (Beltagy et al., 2019)	81.90	77.50	84.30
	HKG Asada et al. (2023)	85.40	-	-
Large language model-based biomedical models	GPT-3.5-turbo (Zhang et al., 2024b)	-	76.80	60.20
	GPT-3.5 (Chen et al., 2025)	34.40	61.91	-
	BioGPT (Luo et al., 2022)	40.76	-	-
	GPT-4 (Chen et al., 2025)	65.58	65.43	-
	SciFive (Phan et al., 2021)	83.67	78.00	-
Task-specific optimized models	HEFed (Wang et al., 2025)	-	-	84.45
	BioM-ELECTRA$_{large}$ (Alrowili et al., 2021)	-	78.60	-
	TaMM-BERT$_{large}$ (Chen et al., 2021)	71.25	67.50	65.94
	SPBERE (Yang et al., 2023a)	79.20	69.70	-
	BioM-ALBERT$_{xxlarge}$ (Alrowili et al., 2021)	82.04	79.30	-
	NBR (Xu et al., 2023)	84.66	80.54	85.86
	WGCN (Wang et al., 2024)	85.00	81.00	85.14
Ours		**86.58**	**81.28**	**86.87**

Table 3
Ablation study.

Model	DDI	ChemProt	GAD
Ours	86.58	81.28	86.87
w/o NLI	85.45	80.44	85.57
w/o CDDT	84.62	79.15	84.93
w/o NLI + CDDT	82.13	76.32	82.45
w/o VDM	85.15	80.09	85.53

the reliability and accuracy of the results. These experimental outcomes fully validate the effectiveness of our proposed method.

4.4. Ablation study

To verify the effectiveness of each component of our model, we conducted ablation experiments, with the results presented in Table 3. We further analyzed the performance degradation caused by removing each module.

First, removing the contrastive loss function led to average performance drops of 1.13 %, 0.84 %, and 1.3 % on the DDI, ChemProt, and GAD datasets, respectively. This loss function serves to pull together positive samples while minimizing the similarity to negative samples, thereby explicitly optimizing the model's ability to distinguish between them. For instance, in the DDI dataset, relations such as "drug interaction" and "no interaction" often exhibit subtle semantic differences. Without contrastive learning, the model struggles to differentiate semantically similar samples, resulting in reduced accuracy in entailment score prediction.

Second, when the dual dependency tree (CDDT) construction module was removed, the most significant performance decline occurred on the ChemProt dataset (2.13 %), followed by the DDI (1.96 %) and GAD (1.94 %) datasets. This can be attributed to the finer-grained entity types and more complex long-range dependencies present in ChemProt. Entities in ChemProt are composed of more specific terms, resulting in richer dependency structures. CDDT captures both syntactic and semantic dependencies by integrating two types of dependency trees, allowing the model to learn structural and logical relations within the sentence. In contrast, the DDI dataset uses a masked entity format, which reduces the richness of dependency structures and thus results in a smaller performance impact.

Moreover, to evaluate the synergistic effect of NLI and CDDT, we conducted an ablation experiment in which both modules were removed simultaneously. This led to significant performance drops of 4.45 %, 4.96 %, and 4.42 % on the DDI, ChemProt, and GAD datasets, respectively. The NLI module, optimized through contrastive loss, enhances the model's semantic discrimination, addressing ambiguity in relation classification. The CDDT module, on the other hand, captures complex entity structures and long-range dependencies through dual dependency representations. When both modules are removed, the model loses the ability to parse structural information and distinguish semantic differences, leading to error accumulation and a substantial decline in overall performance.

Lastly, removing the voting decision mechanism selector also led to a decline in the model's performance across all three datasets. The voting decision mechanism introduces a process of multiple decisions, allowing the model to reevaluate and adjust after each judgment, leading to more accurate relation-type determination. This multi-decision process enhances the model's fault tolerance for incorrect predictions, reducing the overall performance decline caused by a single incorrect decision. Specifically, the voting decision mechanism can synthesize multiple judgment results and determine the final prediction result through majority voting, effectively filtering out noise and misjudgments. This not only improves the stability and robustness of the model in complex relation classification tasks but also significantly enhances the overall accuracy of the model.

Through the ablation studies, we demonstrated the effectiveness of the three proposed components: the contrastive learning loss function, the dual dependency trees construction, and the voting decision mechanism selector, all of which play important roles in enhancing the model's performance.

4.5. Analysis

4.5.1. The impact of the dual dependency trees on the model

In this study, we conducted an in-depth exploration of dual dependency trees and compared them with traditional dependency methods

Table 4
Examples of data display from the dataset.

Dataset	Sentence
DDI	These would include a variety of preparations which contain <e1> DRUG </e1> , estrogens, <e2> DRUG </e2> , or glucocorticoids.
ChemProt	<e1> Bevantolol </e1> : a <e2> beta-1 adrenoceptor </e2> antagonist with unique additional actions.

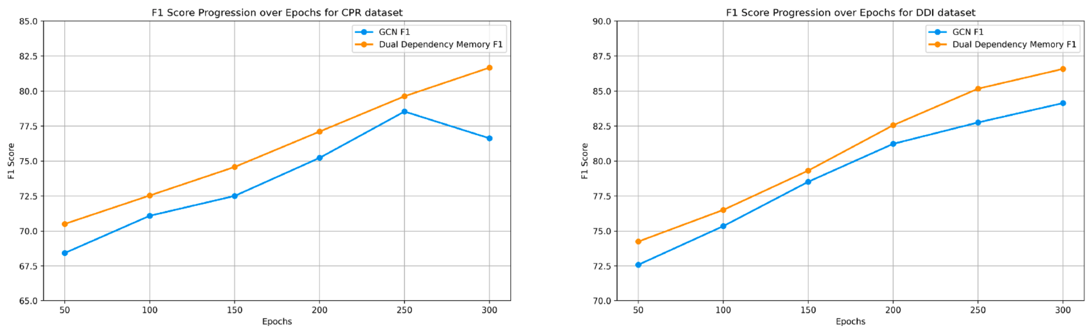

Fig. 4. Comparison results between the dual dependency tree model and the traditional models.

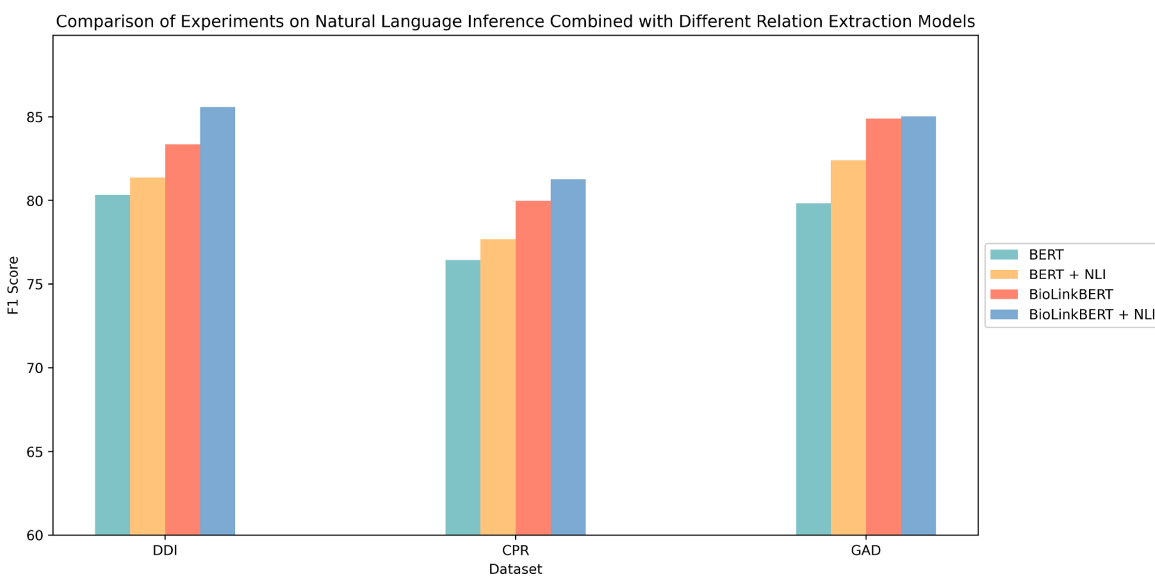

Fig. 5. Experimental results of combining natural language inference with different relation extraction models.

through experimental tests. The experiments were carried out on the DDI and ChemProt datasets. The choice of these two datasets is based on their key differences: entities in the DDI dataset are labeled with uniform identifiers, while in the ChemProt dataset, entities are represented by their full content and consist of multiple words, as shown in Table 4.

To validate the effectiveness of our experiments, we conducted comparative tests using the GCN (Guo et al., 2019b) method and our proposed dual dependency trees on the DDI dataset. As shown in Fig. 4, we found the following: (1) In the DDI dataset, both the traditional method and our dual dependency approach showed continuous improvement in F1 scores as training epochs increased. This indicates that the predictive performance of the models improved with deeper training on the dataset. Furthermore, our dual dependency approach almost consistently outperformed the GCN throughout the training period, further demonstrating the efficacy of our method. (2) In terms of performance, due to the entity type structure in the DDI dataset resembling a masked form, our method obtained less contextual information. Therefore, the performance gap between the traditional method and ours was not significant for the DDI dataset. However, the overall performance still highlighted the advantages of our dual dependency trees, showing more stable and substantial growth in performance. (3) The potential benefit of the dual dependency approach lies in its ability to extract more contextual information and dependencies between entities, which enables the model to achieve higher prediction accuracy when dealing with complex drug interactions. Meanwhile, although the performance of the GCN also gradually improved with training, its growth rate and final performance did not reach the level of the dual dependency approach, indicating that our method is better suited for handling such complex data.

In the ChemProt dataset, the overall performance of the dual dependency approach was superior to that in the DDI dataset. This is because the entities in the ChemProt dataset consist of specific words, allowing our method to better capture contextual information for analysis. Additionally, we observed that the GCN method showed signs of overfitting at 250 epochs while our dual dependency approach continued to maintain good performance. This demonstrates that the dual dependency method performs well regardless of whether the entity content is specific or not.

4.5.2. Contribution of natural language inference to relation extraction models

In the task of relation extraction, we believe that NLI models can indirectly provide additional contextual information and implicit relation recognition capabilities to relation extraction models. Based on this assumption, we combined the traditional relation extraction model BERT

(Baldini Soares et al., 2019; Wang et al., 2019; Wu et al., 2018) and its variant in the biomedical field, BioLink-BERT (Yasunaga et al., 2022), incorporating the NLI model for relation extraction. As shown in Fig. 5, Through experiments and analysis, we reached the following key conclusions: (1) The combination of biomedical relation extraction models and NLI methods performs optimally across multiple datasets: We conducted comparative experiments on the DDI, ChemProt, and GAD datasets, and the results showed that the BioLink-BERT model, combined with NLI, outperformed other models on all three datasets. This indicates that leveraging relation extraction models specifically designed for the biomedical domain, in combination with NLI, can effectively handle complex biomedical texts and specialized terminology, leading to a significant improvement in model performance. (2) The enhancement effect of NLI on relation extraction is significant: By comparing models with and without NLI, we found that NLI can significantly improve the performance of the models. In all datasets, models combined with NLI achieved higher F1 scores than those without NLI. This suggests that cross-domain models can enhance their ability to recognize complex relations by providing additional semantic reasoning capabilities. (3) The synergy between biomedical relation extraction models and NLI is evident: Compared to basic relation extraction models, the domain-specific BioLink-BERT model exhibited stronger performance, particularly when combined with NLI, further boosting its effectiveness. This result not only demonstrates the validity of using domain-specific models but also highlights the contribution of NLI to relation extraction tasks, especially when inferring implicit relations where NLI plays a crucial role.

5. Conclusion

In this study, we demonstrated that incorporating word dependency extraction enhances cross-domain model performance, surpassing state-of-the-art results across multiple datasets. By leveraging word dependencies, our approach improves the contextual understanding of Natural Language Inference models, benefiting biomedical Relation Extraction through cross-task learning.

To achieve this, we proposed a novel method that transforms relations into natural language hypotheses and utilizes dependency analysis to capture semantic structures. Additionally, contrastive learning with an infoNCE loss function enhances prediction accuracy, while a voting mechanism improves reliability.

Experiments on three biomedical RE benchmarks validate our approach's effectiveness and generalizability. Future work will explore integrating large language models for dynamic dependency tree generation and adaptive prompt design to further refine performance and applicability.

CRediT authorship contribution statement

Xueying Li: Conceptualization, Methodology, Validation, Data curation, Writing – original draft; **Di Zhao:** Resources, Writing – review & editing, Project administration, Funding acquisition; **Jiana Meng:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition; **Hongfei Lin:** Supervision.

Statements of ethical approval

This is not required for this study because no human or animal directly participated in this study.

Data availability

The code and data have been highlighted in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Di Zhao reports financial support was provided by Fundamental Research Funds for the Central Universities. Di Zhao reports a relationship with Fundamental Research Funds for the Central Universities that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Fundamental Research Funds for the Central Universities (Grant No. 0854-53).

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